HUM: Hear, Understand, Match for Unseen Audio-Text Alignment

Afeef Ali (Team Leader)
PID: aali11
Dept. of Computer Science
Virginia Tech
aali11@vt.edu

Kean Jaldin Guzman
PID: keanjg28
Dept. of Computer Science
Virginia Tech
keanjg28@vt.edu

Erdem Yilmaz
PID: erdemy
Dept. of Computer Science
Virginia Tech
erdemy@vt.edu

Anthony Merino
PID: merino
Dept. of Computer Science
Virginia Tech
merino@vt.edu

Project web page: https://aliai11.github.io/HUM/

***Abstract*—Automatic metrics for text-to-audio generation still agree poorly with human judgments of whether a clip matches its prompt. The ICASSP 2027 XACLE Grand Challenge asks for a model that predicts the mean human alignment score for an (audio, prompt) pair and tests on audio from generators never seen in training. We introduce HUM (Hear, Understand, Match), which fine-tunes the Qwen2.5-Omni audio-language model as a yes/no verifier and fuses it with frozen CLAP embeddings. At Checkpoint II, the fine-tuned 7B verifier alone reaches test SRCC 0.703, against 0.565 for the official baseline and 0.616 for our best frozen-embedding model, a gated fusion of M2D-CLAP and Human-CLAP. Under leave-one-generator-out evaluation on three hidden generators, it raises held-out-generator SRCC from 0.35 to 0.63 on test, with every paired 95% bootstrap interval above zero, and narrows the seen-to-unseen gap from 0.21 to 0.07. Three cheaper fixes on frozen embeddings, a within-prompt ranking loss, listener-standardized targets, and a gradient-reversal generator classifier, do not improve unseen-generator SRCC. Next, we combine the verifier and the CLAP experts in one network and complete all eight generator folds.**

***Index Terms*—Text-to-audio evaluation, audio-language models, learning to rank, domain generalization, parameter-efficient fine-tuning.**

I. Introduction

A. Motivation

Text-to-audio (TTA) models such as AudioLDM and Tango can turn a sentence into sound, but checking whether the sound actually matches the sentence still requires human listening tests, which are slow, costly, and hard to reproduce. The standard automatic proxy, CLAPScore, correlates weakly with human relevance ratings: on the RELATE dataset, LAION-CLAP reaches a Spearman rank correlation (SRCC) of only 0.280 [1], [2], and contrastive audio-text models struggle with event order and attribute binding [3]. A learned evaluator that agrees with listeners would make TTA benchmarking cheaper and could serve as a reward signal for better generators.

B. Problem Statement

Given generated audio a and its prompt t, we predict a score $\hat{y}$ whose ranking over a test set matches the mean human alignment rating $y \in [0, 10]$. XACLE ranks systems by SRCC, with linear correlation (LCC), Kendall's tau (KTAU), and mean-squared error (MSE) as tie-breaks [4]. The 2027 edition adds an *unseen-generation-system* setting: test audio comes from generators that are absent from training [5]. A model can therefore look strong on validation by learning generator-specific artifacts and still fail on the test set, which makes the task a domain-generalization problem rather than only a regression problem.

C. Significance

Every 2026 system optimized correlation under seen generators. The 2026 baseline (BYOL-A + RoBERTa + LSTM) reached 0.3345 test SRCC [4], while the strongest systems reached 0.61–0.64 [6]–[8]. Since new TTA models appear every few months, a practical evaluator must generalize to generators it has never heard. At the same time, a zero-shot audio-language model judge already reaches 0.556 SRCC on RELATE [9], so strong in-domain correlation is an engineering target; generalization to unseen generators is the open research question.

D. Goals and Contributions

We separate engineering targets from the research bet:

- **G1.** Reproduce the official baseline (B1) and our CLAP-regression fallback (B4) by Checkpoint I (Oct. 20).
- **G2.** *Engineering:* reach in-domain validation SRCC at least equal to zero-shot AQAScore [9].
- **G3.** *Research bet:* reduce the leave-one-generator-out gap Δ (Eq. (7)) relative to B4, with a 95% bootstrap interval that excludes zero across at least three folds.
- **G4.** Submit a rule-compliant system to XACLE 2027 by Dec. 3.
- **G5.** Release a reproducible repository, a Gradio demo, and a demonstration video.

Our contributions are (i) a within-prompt ranking objective for generator-invariant alignment scoring, (ii) an unseen-generator benchmark protocol that reports Δ for re-implemented baselines, and (iii) a single-network judge that combines language-model verification with embedding similarity. Sections V and VI report our progress at Checkpoints I and II.

II. Related Work

A. Embedding-Based and Learned Metrics

CLAPScore [10] measures how well audio and text match by comparing their CLAP embeddings. Human-CLAP improves CLAP by fine-tuning it on human ratings [1], while

M2D-CLAP learns stronger general-purpose audio-language features [11]. RELATE released human relevance scores for TTA audio [2], and XACLE 2026 built its benchmark on this lineage [4].

The top XACLE 2026 systems extended these ideas. MoEScore gated several CLAP-based experts together with a BEATs [12]–RoBERTa cross-attention expert, reaching an SRCC of 0.6402 [6]. The third-place TMU system fine-tuned a compact audio-language model using CLAP pseudo-labels and a listwise ListNet loss [13], reaching 0.632 [7]. ListNet ranks clips across the whole batch regardless of prompt; HUM adds a ranking loss restricted to clips of the same prompt, so the model learns which clip best matches a prompt rather than which prompts are easier to score. SPO-CLAPScore standardized each listener's ratings, reaching 0.6142 [8].

HUM borrows gating, ranking, and listener standardization from these systems. Unlike them, HUM also targets generalization to audio-generation systems not seen during training, which is a problem none of these systems needed to solve, since XACLE 2026 tested only on seen generators.

B. Audio-Language Models as Judges

AQAScore asks Qwen2.5-Omni [14] whether a clip contains the events described in its prompt, using the probability of a "Yes" answer as the score; zero-shot, it reaches SRCC 0.556 on RELATE, above similarity-based metrics [9]. AuroLA repurposes Qwen2.5-Omni-7B for audio-text retrieval and adds a yes/no reranker [15]. HUM fine-tunes such a verifier with LoRA [16] (QLoRA [17] on smaller GPUs), rather than using it zero-shot, and fuses its output with an embedding-based signal instead of relying on the language model alone.

C. Generalization to Unseen Systems

Learned evaluators can overfit to generator-specific acoustic cues instead of semantic alignment [18]; in AudioMOS 2025, for example, systems degraded on unseen synthesis engines and vocoders [19]. Domain generalization methods counter this bias: domain-adversarial training with a Gradient Reversal Layer (GRL) [20] trains the model to score alignment without recognizing the source generator, and test-time adaptation methods like Tent [21] adjust predictions on unlabeled test batches. HUM's primary invariance signal, however, is the within-prompt ranking loss; the GRL branch and test-time calibration are add-ons that are kept only if they reduce Δ.

III. Proposed Approach

A. Overview

HUM is one network with three stages (Fig. 1). **Hear:** the clip passes through the frozen Qwen2.5-Omni audio encoder and frozen M2D-CLAP audio and text encoders, whose outputs are computed once and cached. **Understand:** the Qwen2.5-Omni Thinker with trainable LoRA adapters (on a bf16 base, or 4-bit on smaller GPUs; the speech-output Talker is removed), receives the audio tokens and the query "Does this audio contain the sound events described by 't'?" It returns p_{yes}, the normalized probability of "Yes" versus "No," and $\mathbf{h}_s$,

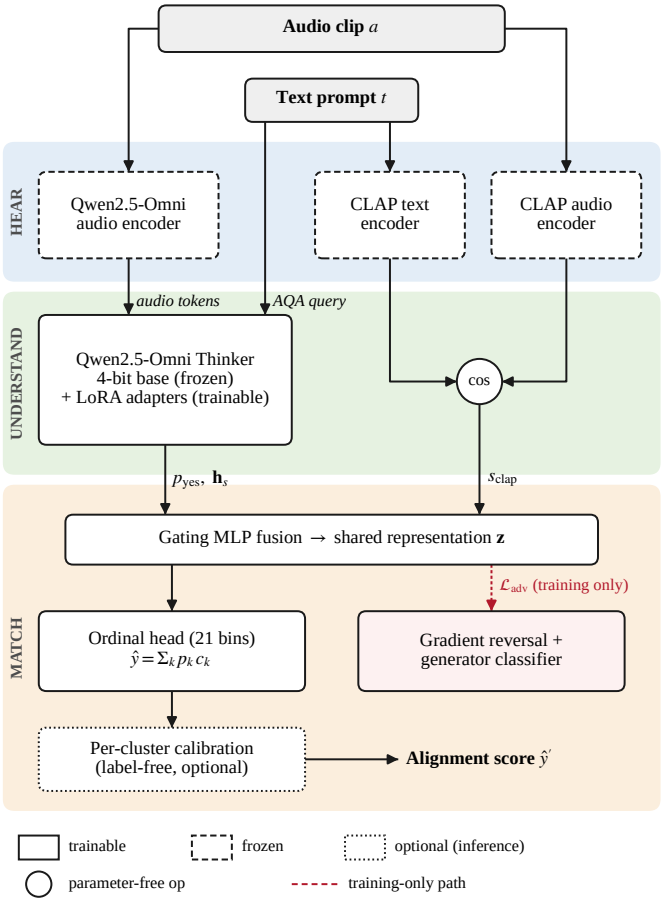


Fig. 1. HUM architecture. *Hear*: frozen audio and CLAP encoders. *Understand*: LoRA-tuned Qwen2.5-Omni verifier. *Match*: gated fusion, ordinal score head, and a training-only gradient-reversal generator classifier. Solid borders denote trainable modules; dashed borders denote frozen modules.

the hidden state at a dedicated score token. **Match:** a gating MLP fuses $\mathbf{h}_s$, p_{yes}, and the CLAP cosine similarity s_{clap} into a shared representation $\mathbf{z}$, and an ordinal head predicts probabilities p_k over $K = 21$ bins $c_k \in \{0, 0.5, \ldots, 10\}$. The score is the expected bin value:

$$\hat{y} = \sum_{k=1}^{K} p_k \, c_k. \tag{1}$$

The 2027 rules prohibit ensembling the results of multiple models but allow internal ensembling within a single model [5]; HUM is one network, so it is compliant.

B. Within-Prompt Ranking

The 2027 development set provides about three clips per prompt, each labeled with the system that generated it (Section IV-B). For clips i and j of the same prompt with human scores $y_i > y_j$, we add a margin ranking loss:

$$\mathcal{L}_{\text{pair}} = \frac{1}{|\mathcal{P}|} \sum_{(i,j)\in\mathcal{P}} \max\bigl(0,\ m - (\hat{y}_i - \hat{y}_j)\bigr), \tag{2}$$

where $\mathcal{P}$ is the set of within-prompt pairs whose score difference exceeds a noise threshold and m is a margin. Because

both clips share the prompt, prompt difficulty cancels, and the model must rank clips by how well each one realizes the same text. We hypothesize that this reduces reliance on generator artifacts without requiring generator labels. Mini-batches are built so that all clips of a prompt appear together.

C. Generator Invariance

To filter generator-specific information out of $\mathbf{z}$, HUM adds an adversarial branch. During training, $\mathbf{z}$ passes through a GRL [20] into a two-layer MLP that predicts the source generator:

$$\mathcal{R}(\mathbf{z}) = \mathbf{z}, \qquad \frac{\partial \mathcal{R}(\mathbf{z})}{\partial \mathbf{z}} = -\mathbf{I}. \tag{3}$$

The 2027 release includes generator IDs, so the branch trains with a standard cross-entropy loss $\mathcal{L}_{\text{adv}}$ and no pseudo-domain clustering is needed. To keep adversarial gradients from destabilizing early training, its weight $\gamma(\tau)$ ramps from 0 to $\gamma_{\max}$ as in (5).

D. Training Objective

The full objective is

$$\mathcal{L} = \mathcal{L}_{\text{rank}} + \alpha\,\mathcal{L}_{\text{ord}} + \beta\,\mathcal{L}_{\text{mse}} + \delta\,\mathcal{L}_{\text{pair}} + \gamma(\tau)\,\mathcal{L}_{\text{adv}} \tag{4}$$

where $\mathcal{L}_{\text{rank}}$ is ListNet over the mini-batch [13], which directly rewards correct orderings and matches the rank-based SRCC; $\mathcal{L}_{\text{ord}}$ is the KL divergence from a Gaussian-softened target over the 21 bins ($\sigma = 0.5$), a soft-bin strategy used by top AudioMOS 2025 systems [19]; and $\mathcal{L}_{\text{mse}}$ on $\hat{y}$ stabilizes the score scale for the LCC and MSE tie-breaks. The weights α, β, δ, and m are tuned on validation data, and the adversarial weight follows

$$\gamma(\tau) = \gamma_{\max}\left(\frac{2}{1 + e^{-10\tau}} - 1\right), \tag{5}$$

where $\tau \in [0, 1]$ is training progress. The default system sets $\gamma_{\max} = 0$; with only eight generators the adversarial signal may be weak, so the GRL branch is kept only if it passes the acceptance rule in Section III-G.

E. Training Stages

S0 caches the frozen audio-encoder and CLAP outputs. **S2**, the default system, fine-tunes on XACLE with $\mathcal{L}_{\text{rank}}$, $\mathcal{L}_{\text{ord}}$, $\mathcal{L}_{\text{mse}}$, and $\mathcal{L}_{\text{pair}}$. Three add-ons run only if they pass the acceptance rule: **S1**, a warm-up on declared external caption–audio pairs labeled by a CLAP teacher, which was the main performance driver for TMU [7]; **S3**, which enables the GRL branch; and per-listener score standardization [8], now possible because listener IDs are released.

F. Optional Test-Time Calibration

Audio from unseen generators can shift scores systematically. Without using test labels, we cluster the evaluation audio representations into C groups with K-means and adjust each score by its cluster's mean prediction μ_c relative to the overall mean μ:

$$\hat{y}' = \hat{y} - \lambda\,(\mu_c - \mu), \qquad \lambda \in [0, 1]. \tag{6}$$

A single monotonic map cannot change SRCC, so the correction must act per cluster. Because full equalization could erase genuine quality differences between generators, λ defaults to 0 and is tuned on validation LOGO folds. The step keeps model weights frozen and is applied only if the rules allow transductive use of the evaluation audio.

G. Baselines, Metrics, and Evaluation Protocol

Baselines: B1 is the official 2027 baseline, a frozen M2D-CLAP encoder with a trained MLP head [22]; B2 is zero-shot CLAPScore; B3 is zero-shot AQAScore [9]; and B4 is the B1 head trained on cached M2D-CLAP embeddings with our losses, which is our guaranteed submission. **Metrics:** we report SRCC (primary), LCC, KTAU, MSE, and per-generator SRCC, with 95% bootstrap intervals from 1,000 prompt-level resamples, because 2026 placements were separated by as little as 0.0001 SRCC [8]. **Protocol:** clips of one prompt never straddle training and evaluation. Leave-one-generator-out (LOGO) cross-validation trains on all but one generator and tests on the held-out one. The headline number is

$$\Delta = \text{SRCC}_{\text{seen}} - \text{SRCC}_{\text{unseen}}. \tag{7}$$

We report Δ for B2–B4 and HUM, which forms our benchmark contribution. **Acceptance rule:** a component is kept only if it lowers Δ without dropping in-domain SRCC below B4. **Ablations:** each loss term removed; with and without S1, GRL, calibration, and listener standardization; zero-shot versus fine-tuned verifier; and 3B versus 7B.

IV. System Design

A. Architecture and Workflow

The pipeline has seven steps: (1) ingest XACLE and declared external data, validate counts and score ranges, and remove RELATE clips that duplicate XACLE audio or prompts; (2) cache frozen features and build prompt-disjoint LOGO folds; (3) train B4, then HUM; (4) score every run with one shared metrics library; (5) select the final configuration with the acceptance rule; (6) run frozen inference on the evaluation release; and (7) export predictions in the official format. Every result comes from a notebook with fixed seeds.

B. Data Sets

Our primary data is the XACLE 2027 development set, described in Section V-A and Table I. RELATE [2] is our external dataset, and AudioCaps and Clotho provide caption–audio pairs for the optional S1 stage; all are on the organizers' pre-approved list.

C. Languages, Tools, and Compute

We build HUM using Python, PyTorch, Hugging Face Transformers and PEFT [16], bitsandbytes [17], torchaudio, scikit-learn, SciPy, Gradio, and GitHub. All compute runs on Google Colab Pro, with notebooks and results shared through one team Drive folder. Because the encoders are frozen and cached, all embedding experiments, including the LOGO runs, run on CPU or an L4 GPU. A100 time is reserved for the 7B

TABLE I
SOURCES IN THE DEVELOPMENT SET

ID	Source	Train†	Val / Test
0	AudioCaps (real recordings)	2,485	1,000 each
1–8	AudioGen, AudioLDM, AudioLDM 2, Make-An-Audio 2, Stable Audio, Tango, Tango 2, TangoFlux	496–747 each	250 each
	Total	7,455	3,000 each

†After removing the 45 prompt-overlap clips.

verifier, which we fine-tune with LoRA in bf16 and resumable checkpoints; one 2-epoch run takes about one A100-hour, well within our budget of about 60. Qwen2.5-Omni (Apache-2.0), including the 3B fallback, is pre-approved.

D. Demo and Usability

A Gradio demo will take an audio file and a prompt and show $\hat{y}$, the 21 bin probabilities, the gating weights, and raw versus calibrated scores; we will report its GPU memory and latency with 4-bit quantization.

E. Risks and Fallbacks

If 7B training is too slow, we fall back to the 3B model or submit B4 alone. If the pairwise loss does not reduce Δ, we report a negative result with confidence intervals, which still supports the benchmark contribution. Section V shows that this risk is real.

V. CHECKPOINT I RESULTS

This section summarizes the frozen-embedding experiments of Checkpoint I; Section VI reports Checkpoint II. Table III collects the in-domain results of both checkpoints.

A. Data and Evaluation Setup

The development set has 7,500/3,000/3,000 training/validation/test pairs of 10-s, 16-kHz clips, rated by four listeners per pair (eight for test) from disjoint listener pools (2,323, 668, and 1,336 listeners). Clips come from real AudioCaps recordings and eight TTA generators (Table I). Our checks confirmed the row counts, that all ratings lie in $[0, 10]$, and that the embedding index matches the metadata. All nine sources appear in all three splits, so the official test split contains no unseen generator. Twelve training prompts also appear in validation (11) or test (1); we remove their 45 training clips in the LOGO runs.

The main results are scored by one shared metrics library that matches the official `evaluate.py` to four decimals and adds prompt-level bootstrap intervals and per-generator SRCC; paired bootstraps are computed in the experiment code. Each LOGO fold holds out one TTA generator from training *and* from best-epoch selection; AudioCaps is never held out because it is real audio. Each fold is run with three seeds for each of the four B4 losses, giving 96 runs.

TABLE II
LOGO RESULTS, MEAN OF 3 SEEDS (BASE: LOSS WITHOUT THE PAIRWISE TERM)

	Unseen SRCC↑		$\Delta\downarrow$	
Held-out generator (test)	Base	$+\mathcal{L}_{\text{pair}}$	Base	$+\mathcal{L}_{\text{pair}}$
Base $\mathcal{L}_{\text{mse}}$, $\delta = 1$				
AudioGen	0.499	0.498	0.061	0.058
AudioLDM	0.431	0.451	0.096	0.071
AudioLDM 2	0.678	0.671	−0.143	−0.141
Make-An-Audio 2	0.373	0.367	0.206	0.202
Stable Audio	0.514	0.535	0.042	0.015
Tango	0.563	0.582	−0.018	−0.044
Tango 2	0.477	0.496	0.093	0.064
TangoFlux	0.227	0.257	0.358	0.313
Mean, test	0.470	**0.482**	0.087	**0.067**
Mean, validation	0.437	**0.447**	0.143	**0.131**
Base $\mathcal{L}_{\text{off}}$, $\delta = 0.03$				
Mean, test	0.469	**0.475**	0.087	**0.081**
Mean, validation	0.433	**0.440**	0.145	**0.141**

B. Experiments

B1 and the embedding cache. The released B1 checkpoint, run with the official scripts, reproduces the published numbers to every reported digit. Since its encoders are frozen, we cached the normalized 768-dimensional M2D-CLAP audio and text embeddings for all 13,500 clips (83 MB instead of 4.2 GB of audio). The B1 head applied to the cache reproduces the released predictions within rounding (maximum difference 0.005).

B4 graders. All use B1's head (linear 1536→1024→256→1, ReLU, dropout 0.3) on the concatenated embeddings, with batches of 12 clips made of whole prompts, so they differ only in the loss: $\mathcal{L}_{\text{mse}}$ or the official loss $\mathcal{L}_{\text{off}}$ (a class-balanced clipped MSE plus a contrastive term [22]), each alone or with $\mathcal{L}_{\text{pair}}$ ($m = 0.5$ points, 0.5-point threshold). $\mathcal{L}_{\text{pair}}$ uses $\delta = 1$ with $\mathcal{L}_{\text{mse}}$; because $\mathcal{L}_{\text{off}}$ is about 12 times smaller than $\mathcal{L}_{\text{mse}}$, its δ is chosen on validation from $\{0.03, 0.1, 0.3, 1\}$. Training uses Adam (learning rate 2×10^{-5}) for 20 epochs, keeps the epoch with the best validation SRCC, and uses seeds 0–2. A code review fixed two bugs in the first version (cross-prompt pairs and uncopied best-epoch weights); all results use the corrected code.

B3 verifiers. Following AQAScore, the score is $10\,p_{\text{yes}}$ from the next-token logits for the query in Section III-A, run in bf16 on one A100 over 498 validation clips from 164 random prompts. AuroLA uses its own reranking prompt.

B2. The score is the cosine similarity s_{clap} of the cached, L_2-normalized M2D-CLAP audio and text embeddings, mapped to the 0–10 scale by a least-squares fit on the training split ($\hat{y} = 29.52\, s_{\text{clap}} - 2.76$); the map changes only MSE.

C. Results

In-domain (Table III): the $\mathcal{L}_{\text{mse}}$ grader slightly exceeds B1 on both splits with very small seed variance, confirming the cache and training pipeline, and $\mathcal{L}_{\text{off}}$ is nearly identical (validation/test SRCC 0.5867/0.5644; paired differences -0.004

TABLE III
IN-DOMAIN RESULTS (ALL GENERATORS SEEN IN TRAINING). TRAINED CLAP-EMBEDDING MODELS: MEAN OF 3 SEEDS; VERIFIER: 1 SEED

	Val	Test			
Model	SRCC↑	SRCC↑	LCC↑	KTAU↑	MSE↓
B2: CLAP similarity	0.427	0.420	0.512	0.289	3.81
B1: official baseline	0.584	0.565	0.614	0.400	3.18
B4: $\mathcal{L}_{\text{mse}}$	0.588	0.567	0.616	0.401	3.22
B4: $\mathcal{L}_{\text{mse}} + \mathcal{L}_{\text{pair}}$ ($\delta = 1$)	0.587	0.558	0.605	0.395	3.24
B4: $\mathcal{L}_{\text{off}}$	0.587	0.564	0.615	0.399	3.19
B4: $\mathcal{L}_{\text{off}} + \mathcal{L}_{\text{pair}}$ ($\delta = 0.03$)	0.589	0.563	0.614	0.398	3.18
B4: $\mathcal{L}_{\text{off}}$, listener-std.	0.584	0.569	0.612	0.402	3.20
Human-CLAP grader ($\mathcal{L}_{\text{off}}$)	0.564	0.517	0.569	0.364	3.50
Gated fusion (M2D + Human)	0.654	0.616	0.677	0.441	2.81
HUM verifier (7B, LoRA)	**0.691**	**0.703**	**0.743**	**0.526**	**2.30**

to $+0.001$). Adding $\mathcal{L}_{\text{pair}}$ to $\mathcal{L}_{\text{mse}}$ ties it on validation (all paired intervals include zero) but lowers test SRCC by 0.008, significantly for two of three seeds. With $\mathcal{L}_{\text{off}}$, $\mathcal{L}_{\text{pair}}$ at $\delta = 1$ dominates the smaller loss (0.5428/0.5324), and $\delta = 0.3$ and 0.1 still hurt (0.5698/0.5450 and 0.5855/0.5585). The chosen $\delta = 0.03$ is neutral (0.5889/0.5631): one seed is significantly better on validation and one significantly worse on test.

Zero-shot verifiers: on the 498-clip subset, Qwen2.5-Omni-7B reaches SRCC 0.495, AuroLA-rerank-7B 0.476, and the trained B1 0.570. We choose Qwen2.5-Omni-7B because it scored higher, although the 0.019 difference is small for 164 prompts. The 0.08 gap to the trained model is what fine-tuning must close.

Unseen generators (Table II): generalization varies widely. TangoFlux and Make-An-Audio 2 are hardest when unseen (0.23 and 0.37), while AudioLDM 2 and Tango are easier than the pooled seen set, giving negative Δ. With $\mathcal{L}_{\text{mse}}$, $\mathcal{L}_{\text{pair}}$ raises unseen SRCC by 0.012 on test and 0.010 on validation and reduces Δ in 7 of 8 test folds and all 8 validation folds, but part of this comes from a lower seen SRCC (0.557 to 0.549 on test). With $\mathcal{L}_{\text{off}}$ at $\delta = 0.03$, the gain is half as large (0.006 on both splits; Δ falls in 6 of 8 folds on each), and seen SRCC is unchanged (0.556 on test). Few paired intervals exclude zero: 4 of 24 test fold-seed runs improve with $\mathcal{L}_{\text{mse}}$, and 2 improve and 1 worsens with $\mathcal{L}_{\text{off}}$, so G3 is not met. Measured against the mean within-generator SRCC instead, the reduction vanishes in both cases ($\leq$0.001).

Interpretation: on frozen embeddings, $\mathcal{L}_{\text{pair}}$ slightly improves the ordering within every generator, seen or unseen, rather than closing the gap. It fails the acceptance rule with $\mathcal{L}_{\text{mse}}$ at $\delta = 1$ but passes narrowly with $\mathcal{L}_{\text{off}}$ at $\delta = 0.03$, so B4 keeps that setting. A frozen-encoder head has little capacity to reorder clips of the same prompt, so we will re-tune δ and test the loss inside the fine-tuned verifier.

VI. CHECKPOINT II PROGRESS REPORT

A. Status of Goals

G1 was completed at Checkpoint I. **G2 is met:** the fine-tuned verifier reaches validation SRCC 0.691, against 0.501 for the same model zero-shot on all 3,000 validation clips. **G3 is partly met:** against B4, the verifier lowers Δ with all three seed-wise intervals above zero in two of the three hidden-generator folds run so far (TangoFlux and Make-An-Audio 2, test split), and G3 asks for three; its gain in unseen-generator SRCC is significant in all 18 comparisons. **G4** is on schedule, with the verifier as a submission candidate and B4 as the fallback. **G5:** the repository and web page are live; the demo has not started.

B. Changes Since Checkpoint I

- The 7B verifier trains with LoRA on a bf16 base on an 80 GB A100, so the 4-bit base and the 3B fallback were not needed.
- The verifier is trained alone first, as a soft yes/no judge (Section VI-C); the gating fusion and ordinal head come next.
- Human-CLAP with gated fusion, listener standardization, and the GRL branch (S3) were first tested on frozen embeddings, where each run costs minutes.
- Next to Δ we now report unseen-generator SRCC, because Δ can also fall when seen SRCC drops (Section VI-F).

C. Fine-Tuning the AI Listener

The verifier, our "AI listener," is the Qwen2.5-Omni-7B Thinker fine-tuned with LoRA (rank 16, $\alpha = 32$, dropout 0.05) on the attention and MLP projections of its language model; the audio and vision encoders stay frozen. Each clip is paired with the query of Section III-A, and we read the margin $s = \ell_{\text{Yes}} - \ell_{\text{No}}$ between the two answer logits at the last position, so $p_{\text{yes}} = \sigma(s)$ and $\hat{y} = 10\, p_{\text{yes}}$. The loss is binary cross-entropy between p_{yes} and the soft target $y/10$. We train for 2 epochs with AdamW (learning rate 10^{-4}, 3% warm-up, cosine decay) on batches of 6 clips made of whole prompts with 3 accumulation steps, check validation SRCC every half epoch, and keep the best checkpoint; a checkpoint every 150 updates lets a run resume after a disconnect. A run takes about one A100-hour (9 Colab compute units). The in-domain run uses all 7,500 training clips, like B4. Each hidden-generator run follows the LOGO rules of Section V-A, removing the generator from training and from checkpoint selection. We ran TangoFlux, Make-An-Audio 2, and AudioLDM, the three generators on which B4 is weakest.

D. Add-Ons on Frozen Embeddings

Human-CLAP and gated fusion. Notebook 11 caches the 512-dimensional, L_2-normalized Human-CLAP audio and text embeddings [1] for all 13,500 clips, in the same row order as the M2D-CLAP cache. Notebook 12 trains the B4 head on the Human-CLAP embeddings alone, and a gated fusion with one head per encoder: a gating MLP (64 hidden units, sigmoid output) reads both encoders' embeddings and the two cosine similarities and outputs a weight g on the Human-CLAP head, so that $\hat{y} = g\,\hat{y}_{\text{H}} + (1 - g)\,\hat{y}_{\text{M}}$. The fusion is trained end to end with $\mathcal{L}_{\text{off}}$ on the fused output over seeds 0–2, with the batching, optimizer, and best-epoch rule of B4.

Listener standardization. Some listeners rate harshly and others generously, so a clip's mean rating depends partly on

TABLE IV
HIDDEN-GENERATOR RESULTS: B4 ($\mathcal{L}_{\text{off}}$, MEAN OF 3 SEEDS) VS. THE FINE-TUNED VERIFIER (1 SEED)

		Unseen SRCC↑			Δ ↓	
Hidden generator	Split	B4	HUM	Gain [low]	B4	HUM
TangoFlux	Test	0.240	**0.537**	+0.30 [0.16]	0.343	**0.165***
	Val	0.257	**0.437**	+0.18 [0.04]	0.324	**0.250**
Make-An-Audio 2	Test	0.380	**0.690**	+0.31 [0.20]	0.196	**0.011***
	Val	0.343	**0.522**	+0.18 [0.07]	0.256	**0.175**
AudioLDM	Test	0.429	**0.648**	+0.22 [0.10]	0.098	**0.038**
	Val	0.389	**0.551**	+0.16 [0.05]	0.185	**0.129**
Mean	Test	0.350	**0.625**	+0.28	0.212	**0.071**
	Val	0.330	**0.503**	+0.17	0.255	**0.185**

[low]: smallest lower 95% bound of the paired gain over the 3 B4 seeds. *Δ reduction significant for all 3 seeds.

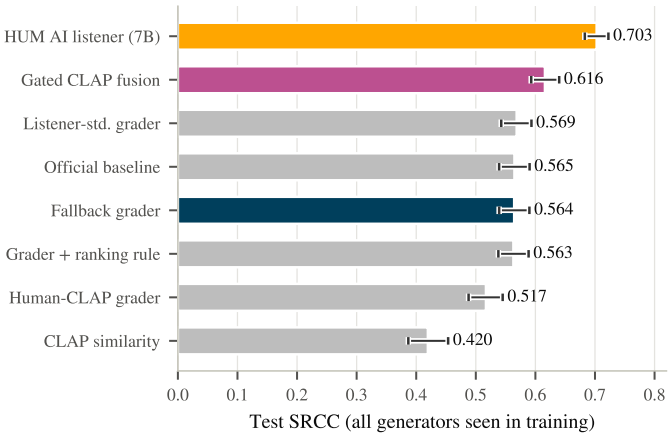


Fig. 2. In-domain test SRCC. Bars: mean over seeds; whiskers: 95% prompt-level bootstrap intervals. Colors mark B4, the fallback grader (dark blue), the gated CLAP fusion (magenta), and the fine-tuned verifier (orange); other models are gray.

who rated it. For each of the 2,323 training listeners l, with N_l ratings of mean M_l and variance V_l, we shrink the mean and standard deviation toward the global values μ_g and σ_g with weight $K = 5$ and floor the deviation at 0.5:

$$\mu_l = \frac{N_l M_l + K\mu_g}{N_l + K}, \quad \sigma_l = \max\Big(0.5, \sqrt{\tfrac{(N_l-1)V_l + K\sigma_g^2}{N_l - 1 + K}}\Big). \tag{8}$$

Each rating r becomes $z = (r - \mu_l)/\sigma_l$, and the new target of a clip is $\mu_g + \sigma_g \bar{z}$, clipped to $[0, 10]$, where $\bar{z}$ is the mean z over the clip's listeners. We retrain B4 ($\mathcal{L}_{\text{off}}$) on these targets with seeds 0–2 and otherwise unchanged training and, as for every model, score validation and test against the official mean ratings.

Generator-adversarial branch (S3). Notebook 13 attaches the GRL branch to the 256-dimensional penultimate representation $\mathbf{z}$ of the B4 grader ($\mathcal{L}_{\text{off}} + 0.03\mathcal{L}_{\text{pair}}$): a two-layer classifier ($256 \to 128 \to 9$) predicts the source generator with cross-entropy, and its weight ramps as in (5). We tried $\gamma_{\max} \in \{0.01, 0.03, 0.1\}$ in-domain and on all 8 LOGO folds, with 3 seeds each (81 runs), and chose $\gamma_{\max}$ by validation unseen SRCC, which picked the smallest value (0.445, against 0.439 and 0.406).

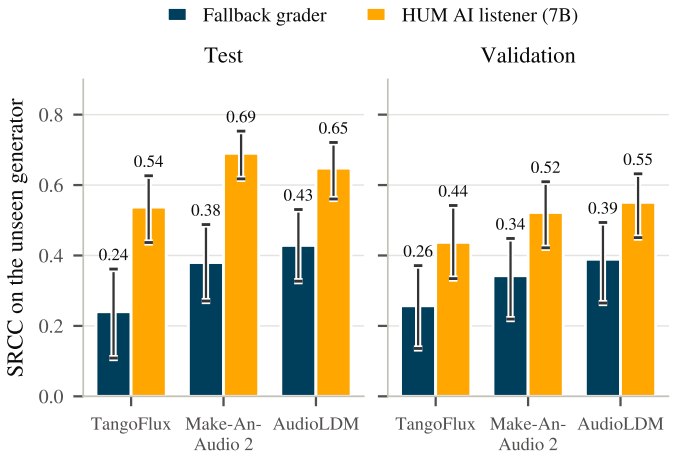


Fig. 3. SRCC on the hidden generator in three LOGO folds (fallback grader B4: mean of 3 seeds; verifier: 1 seed). Whiskers: 95% prompt-level bootstrap intervals over the 250 hidden-generator clips.

E. Results

In-domain (Table III, Fig. 2): fine-tuning lifts the verifier from validation SRCC 0.501 zero-shot to 0.674 after half an epoch and 0.691 after two, where it levels off. On test it reaches 0.703 [0.683, 0.723], 0.14 above B1 and 0.09 above the gated fusion, and it is best on every metric, including MSE (2.30 against 3.18 for B1). Test exceeds validation because test clips have eight ratings instead of four. For context, the strongest 2026 systems reported 0.61–0.64 on a different test set.

Unseen generators (Table IV, Fig. 3): with the generator hidden from training, the verifier raises unseen SRCC on test from 0.240 to 0.537 (TangoFlux), 0.380 to 0.690 (Make-An-Audio 2), and 0.429 to 0.648 (AudioLDM), and by 0.16–0.18 on validation. A paired prompt-level bootstrap against each B4 seed puts all 18 gains (3 folds × 2 splits × 3 seeds) above zero. Because the verifier also raises seen SRCC by 0.10–0.16, Δ falls less: on test from 0.212 to 0.071 on average, significantly for TangoFlux and Make-An-Audio 2 but not AudioLDM; on validation it falls in all three folds, but every interval includes zero. These are three of eight folds and one verifier seed.

Add-ons on frozen embeddings. The gated fusion of M2D-CLAP and Human-CLAP raises test SRCC from 0.564 to 0.616, significantly for all 3 seeds on both splits, although Human-CLAP alone is worse (0.517). A fixed 50/50 average of the two graders, a diagnostic only since ensembling separate models is not allowed, reaches 0.582, so the trained gate adds 0.034, yet its weight on Human-CLAP stays between 0.41 and 0.54 for every generator. The fusion has not been run under LOGO, and an overlap between RELATE, on which Human-CLAP was fine-tuned, and the XACLE clips is not yet ruled out. Zero-shot, the Human-CLAP cosine also beats the M2D-CLAP cosine on validation (SRCC 0.503 against 0.427) but not on test (0.423 against 0.420), a gap we cannot yet explain, so the fusion gains should be read with caution until the overlap check is done. Listener-standardized targets move each training target by 0.45 points on average but leave SRCC unchanged (test $+0.004$, validation -0.003; no seed

significant either way). Possibly the shifts barely reorder the training targets, or the frozen-embedding head cannot exploit them. The GRL branch at $\gamma_{\max} = 0.01$ leaves unseen SRCC unchanged (0.475 to 0.474 on test, mean of 24 fold-seed runs). It lowers Δ from 0.081 to 0.064, in 7 of 8 test folds and significantly in 11 of 24 runs, but only by lowering seen SRCC (0.556 to 0.538); stronger settings lower both. With the M2D-CLAP encoder frozen, the head can likely remove generator cues from $\mathbf{z}$ only by also discarding cues it needs for scoring. Listener standardization and GRL therefore fail the acceptance rule and are not kept.

F. Findings

(i) *The verifier is where the gains are.* Fine-tuning the audio-language model adds 0.14 test SRCC in-domain and 0.28 on unseen generators, while no change to the frozen-embedding grader that we ran under LOGO adds more than 0.012 unseen SRCC. (ii) *The unseen-generator gap is largely a property of frozen embedding features:* on these three hidden generators B4 loses 0.21 SRCC between seen and unseen audio, the verifier 0.07. (iii) Δ *alone can mislead.* GRL shrinks Δ by making seen generators worse, whereas the verifier shrinks it less because it improves seen generators too, so we report unseen SRCC next to Δ and keep the in-domain condition of the acceptance rule. (iv) *Training is cheap.* Most of the gain arrives in the first half epoch and a run costs about one A100-hour, so the remaining folds, seeds, and the full HUM network fit our compute budget.

G. Progress by Team Member

Afeef Ali (leader). *Done:* built the 7B LoRA verifier and its unattended, resumable training notebook, and ran the smoke test, the in-domain run, and three hidden-generator folds (about 4 A100-hours); reviewed the teammates' notebooks, correcting a doubly applied GRL strength and the strength-selection criterion in notebook 13; helped set up notebook 14 (listener standardization, report tables, and figures); coordinated the team and drafted this report. *Next:* the full HUM network (verifier, CLAP experts, gate, and ordinal head), the remaining folds and seeds, and the submission pipeline.

Kean Jaldin Guzman. *Done:* cached Human-CLAP embeddings for all 13,500 clips (notebook 11); trained the Human-CLAP grader and the gated fusion over three seeds, with paired bootstraps, per-generator SRCC, and gate-weight analysis (notebook 12); maintained the shared metrics and bootstrap library. *Next:* the RELATE overlap check, the gated fusion under LOGO (8 folds $\times$ 3 seeds), and a concatenated-features control to test whether the gate itself matters.

Erdem Yilmaz. *Done:* implemented the generator-adversarial branch (S3) with the $\gamma(\tau)$ schedule on the B4 grader; ran the $\gamma_{\max}$ sweep over all 8 LOGO folds and 3 seeds (81 runs, notebook 13) and selected $\gamma_{\max} = 0.01$ on validation unseen SRCC; ran paired prompt-level bootstraps for every GRL run; and scored the verifier's three hidden-generator folds against each B4 seed. *Next:* score the remaining five verifier folds as they finish, test the cluster calibration of (6) with λ tuned on LOGO validation splits, and help freeze the submission checkpoints.

Anthony Merino. *Done:* implemented listener standardization with Bayesian-smoothed per-listener z-scores (8), retrained the B4 grader on the standardized targets over three seeds, and scored every run against the official mean ratings; compared these runs with B4, the gated fusion, and the verifier using paired bootstraps; and produced the report tables and figures (notebook 14). *Next:* the remaining ablations, the method freeze, and the dry-run submission on validation.

H. Next Steps

Before the evaluation release on Nov. 19 we will (1) run the verifier on the remaining five hidden-generator folds and two more in-domain seeds (about seven A100-hours), which settles G3; (2) build the full HUM network, feeding the verifier's p_{yes} and $\mathbf{h}_s$ together with the M2D-CLAP and Human-CLAP embeddings into the gate and ordinal head, trained jointly, and keep it only if it beats the verifier alone under LOGO; (3) re-test $\mathcal{L}_{\text{pair}}$ inside the verifier; and (4) check the RELATE overlap before relying on Human-CLAP. We will freeze the method by Nov. 18, score the evaluation set, and submit up to four systems by Dec. 3: HUM, the verifier alone, the gated fusion, and B4.

ACKNOWLEDGMENT

AI tools were used to help draft and edit this report and to assist with coding; all technical choices, results, claims, and citations were reviewed by the team.

REFERENCES

[1] T. Takano *et al.*, "Human-CLAP: Human-perception-based contrastive language-audio pretraining," in *Proc. APSIPA ASC*, 2025.

[2] Y. Kanamori, Y. Okamoto, T. Takano, S. Takamichi, Y. Saito, and H. Saruwatari, "RELATE: Subjective evaluation dataset for automatic evaluation of relevance between text and audio," in *Proc. Interspeech*, 2025, pp. 3155–3159.

[3] S. Ghosh *et al.*, "CompA: Addressing the gap in compositional reasoning in audio-language models," in *Proc. Int. Conf. Learn. Represent. (ICLR)*, 2024.

[4] Y. Okamoto, R. Takizawa, M. Kishi, Y. Kanamori, N. Tonami, R. Nagase, S. Takamichi, and K. Imoto, "XACLE Challenge 2026: The first x-to-audio alignment challenge," in *Proc. IEEE Int. Conf. Acoust., Speech, Signal Process. (ICASSP)*, 2026, pp. 21877–21879.

[5] XACLE Organizers, "XACLE Challenge 2027," ICASSP 2027 Signal Processing Grand Challenge, 2026. [Online]. Available: https://xacle.org/2027/

[6] B. Sun, Y. Xiao, and H. Yin, "MoEScore: Mixture-of-experts-based text-audio relevance score prediction for text-to-audio system evaluation," in *Proc. IEEE ICASSP*, 2026, arXiv:2601.06829.

[7] A. Tsutsumi, K. Tanaka, and S. Shiota, "The TMU system for the XACLE challenge: Training large audio language models with CLAP pseudo-labels," in *Proc. IEEE ICASSP*, 2026, arXiv:2602.00604.

[8] T. Takano *et al.*, "SPO-CLAPScore: Enhancing CLAP-based alignment prediction system with standardize preference optimization, for the first XACLE challenge," arXiv:2601.02900, 2026.

[9] C.-Y. Kuan, K.-W. Chang, and H.-Y. Lee, "AQAScore: Evaluating semantic alignment in text-to-audio generation via audio question answering," arXiv:2601.14728, 2026.

[10] Y. Wu, K. Chen, T. Zhang, Y. Hui, T. Berg-Kirkpatrick, and S. Dubnov, "Large-scale contrastive language-audio pretraining with feature fusion and keyword-to-caption augmentation," in *Proc. IEEE ICASSP*, 2023.

[11] D. Niizumi, D. Takeuchi, M. Yasuda, B. T. Nguyen, Y. Ohishi, and N. Harada, "M2D-CLAP: Exploring general-purpose audio-language representations beyond CLAP," *IEEE Access*, vol. 13, 2025.

[12] S. Chen, Y. Wu, C. Wang, S. Liu, D. Tompkins, Z. Chen, and F. Wei, "BEATs: Audio pre-training with acoustic tokenizers," in *Proc. Int. Conf. Mach. Learn. (ICML)*, 2023, pp. 5178–5193.
[13] Z. Cao, T. Qin, T.-Y. Liu, M.-F. Tsai, and H. Li, "Learning to rank: From pairwise approach to listwise approach," in *Proc. ICML*, 2007, pp. 129–136.
[14] J. Xu *et al.*, "Qwen2.5-Omni technical report," arXiv:2503.20215, 2025.
[15] J. Xu, C. Thomé, D. Horak, W. Xie, and A. Zisserman, "Scaling audio-text retrieval with multimodal large language models," arXiv:2602.18010, 2026.
[16] E. J. Hu *et al.*, "LoRA: Low-rank adaptation of large language models," in *Proc. ICLR*, 2022.
[17] T. Dettmers, A. Pagnoni, A. Holtzman, and L. Zettlemoyer, "QLoRA: Efficient finetuning of quantized LLMs," in *Proc. Adv. Neural Inf. Process. Syst. (NeurIPS)*, 2023.
[18] K.-T. Huang, C.-C. Wang, C.-Y. Yang, H.-S. Lee, H.-M. Wang, and B. Chen, "Robust generative audio quality assessment: Disentangling quality from spurious correlations," arXiv:2603.16201, 2026.
[19] W.-C. Huang *et al.*, "The AudioMOS Challenge 2025," in *Proc. IEEE Autom. Speech Recognit. Understanding Workshop (ASRU)*, 2025, arXiv:2509.01336.
[20] Y. Ganin, E. Ustinova, H. Ajakan, P. Germain, H. Larochelle, F. Laviolette, M. Marchand, and V. Lempitsky, "Domain-adversarial training of neural networks," *J. Mach. Learn. Res.*, vol. 17, no. 59, pp. 1–35, 2016.
[21] D. Wang, E. Shelhamer, S. Liu, B. Olshausen, and T. Darrell, "Tent: Fully test-time adaptation by entropy minimization," in *Proc. ICLR*, 2021.
[22] XACLE Organizers, "XACLE 2027 baseline model," GitHub, 2026. [Online]. Available: https://github.com/XACLE-Challenge/XACLE_2027_baselinemodel

APPENDIX A
TASK ASSIGNMENT FOR EACH MEMBER

TABLE V
TASK ASSIGNMENT FOR EACH MEMBER (UPDATED AT CHECKPOINT II)

Member	**Primary Responsibilities**
[Leader] Afeef Ali	Leading overall coordination, data download, and the frozen-feature cache. Fine-tuning the Qwen2.5-Omni-7B verifier with LoRA (in-domain and hidden-generator runs) and building the full HUM network: the score token, gating fusion with the CLAP experts, the ordinal head, the Eq. (4) loss function, and final inference.
Anthony Merino	Building the CLAP-regression fallback (B4) and its loss-weight sweep, listener standardization, the results tables and report figures, the project web page, and the optional S1 pseudo-label warm-up.
Erdem Yilmaz	Generator analysis and the LOGO protocol, the GRL head with its γ schedule for Stage S3, paired-bootstrap analysis of all hidden-generator runs, and per-cluster calibration with λ tuning.
Kean Jaldin Guzman	Managing data ingestion, validation, and RELATE de-duplication. Maintaining the metrics and bootstrap library, building the Human-CLAP cache and the gated fusion of the CLAP experts, and formatting the final submission.
All	Weekly syncs, code review of every notebook, and presentations.

APPENDIX B
SCHEDULE

TABLE VI
PROJECT SCHEDULE (FALL 2026), WITH STATUS AT CHECKPOINT II

Week (2026)	**Tasks**	**Milestone / Deliverable**	**Status**
Sep 15–22	Form team, elect leader; survey XACLE 2026 papers; draft proposal; set up repo and web page	**Sep 22: Proposal due**	Done
Sep 23–29	Sept. 24 data + baseline release: download; verify rules, metric, generator/listener IDs; data validation	Rules/data checklist	Done
Sep 30–Oct 6	Run official baseline (B1), zero-shot CLAPScore (B2) and AQAScore (B3); cache CLAP features; build LOGO folds	Baseline table	Done by Oct 19
Oct 7–19	Train fallback B4; first LOGO Δ numbers; zero-shot verifier comparison; official-loss graders and $\mathcal{L}_{\text{pair}}$ weight sweep	**Oct 20: Checkpoint I** (G1)	Done; first QLoRA run moved to Oct 20
Oct 20–Nov 2	First QLoRA run of HUM (S2; 7B, or 3B if no 7B epoch by Oct 26); $\mathcal{L}_{\text{pair}}$ weight in the verifier; Human-CLAP cache and gating fusion; GRL head (S3); optional S1	Fused model on LOGO	Done: 7B verifier (S2, verifier only), Human-CLAP and gated fusion, GRL (S3) on B4; S1 and $\mathcal{L}_{\text{pair}}$ in the verifier moved
Nov 3–9	The first five ablations; calibration prototype with λ tuning; demo prototype	**Nov 10: Checkpoint II + 15-min talk**	Done: three verifier folds, GRL sweep, listener standardization; calibration and demo moved
Nov 10–18	Full HUM network; remaining verifier folds and seeds; $\mathcal{L}_{\text{pair}}$ in the verifier; RELATE overlap check; remaining ablations; freeze method; dry-run submission on validation	Frozen configuration	Planned
Nov 19–Dec 2	Nov. 19 evaluation release: frozen inference, optional calibration, sanity checks; draft system description	Final predictions	Planned
Dec 1–8	Submit to XACLE (Dec. 3); final presentation (Dec. 1/3/8); record demo video	**Dec 3: XACLE submission**	Planned
Dec 9–11	Final report, repo tag, video upload, peer evaluations	**Dec 11 (8 PM): Final report**	Planned

APPENDIX C
ACKNOWLEDGMENT

This report was prepared with the assistance of AI tools, which were used to help generate, edit, and organize content, check references, and assist with code. All technical choices, results, claims, citations, and code are reviewed and verified by the team, and the team takes full responsibility for the work.